

AI/ML FRESHER INTERVIEW EXPERIENCE #3



1. How is Python used in AI field?

Python is the most popular language in AI because it is simple, readable and has a rich ecosystem of libraries. It is used for:

- Data collection and preprocessing (Pandas, NumPy)
- Model building (Scikit-learn, TensorFlow, PyTorch)
- Data visualization (Matplotlib, Seaborn, Plotly)
- NLP tasks (NLTK, SpaCy, Transformers)
- AI application development and automation (APIs, Streamlit, FastAPI, etc.)

2. What is API ?

API (Application Programming Interface) is a set of rules and protocols that allows two software applications to communicate with each other.

In AI, APIs are used to access models (like LLMs), databases, cloud services, or other external tools.

3. What is the key difference between AI and ML?

AI (Artificial Intelligence)	ML (Machine Learning)
• AI is the broader concept of making machines intelligent. • It includes ML, NLP, Robotics, Computer Vision, Expert Systems, etc. • Goal: To create machines that can think and act like humans.	• ML is a subset of AI. • It focuses on algorithms that learn from data and improve performance. • Goal: To learn patterns from data and make predictions or decisions.

4. Can you elaborate Prompt Engineering?

Prompt Engineering is the process of designing and refining inputs (prompts) to get accurate, relevant and consistent responses from an LLM.

It involves:

- Choosing the right instructions and context
- Defining role, format and constraints
- Using techniques like Few-shot, Chain-of-Thought, ReAct, Prompt chaining, etc.

Good prompts = Better outputs.

5. What is LLM and How it influences the output?

LLM (Large Language Model) is a deep learning model trained on massive text data to understand and generate human-like text.

It influences the output through:

- Model size and training data quality
- The prompt (instructions, context, examples)
- Parameters like temperature, top-p, max tokens
- System message and conversation history

6. AI agent vs Agentic AI ?

AI Agent	Agentic AI
• Follows predefined instructions to perform specific tasks. • Works in a single-step or fixed workflow. • Limited memory and decision-making. • Example: FAQ chatbot, rule-based assistant.	• Autonomous, goal-driven and can plan actions. • Has memory, reasoning and can use tools. • Works in multi-step workflows to achieve complex goals. • Example: Auto-research assistant, task planner, self-improving agents.

7. Describe me about Hallucination and how will you overcome through that?

Hallucination is when an LLM generates incorrect, misleading or fabricated information as if it is true.

How to overcome:

- Use RAG (Retrieval Augmented Generation)
- Provide relevant context and references
- Use system prompts to restrict the model
- Validate output with external data / sources
- Lower temperature for factual tasks

8. RAG architecture

RAG (Retrieval Augmented Generation) combines retrieval and generation to give accurate and context-aware answers.



- Retrieves relevant documents from a knowledge base.
- Passes them along with the query to the LLM.
- LLM generates an answer based on the context.

9. What are LangChain and LangGraph?

LangChain: A framework that helps build LLM applications by chaining components like prompts, memory, tools, models and APIs easily.

LangGraph: An extension of LangChain that enables building stateful, multi-step and cyclical workflows using graphs. It is useful for creating advanced AI agents.

10. Asked to build a simple Streamlit application using LLM through API key?

Steps:

1. Get API key from provider (Groq / OpenAI / etc.)
2. Install required libraries: streamlit, requests (or openai)
3. Build UI with streamlit (text_input, button, text_area)
4. Send user prompt to LLM API using the key
5. Display the response in the UI
6. Add error handling and loading indicator

11. How will you explain about Machine learning to a non-technical background student?

Machine Learning is a way to teach computers to learn from examples instead of writing rules for every situation. We give data to the computer and it finds patterns in the data and uses them to make predictions or decisions.

Example: Just like we learn from past experiences, ML models learn from past data.

12. How Tokenization is important?

Tokenization breaks text into smaller pieces called tokens (words, subwords or characters) that the model can understand.

It is important because:

- LLMs cannot understand raw text directly
- It helps in handling unknown words
- It reduces the text into numbers (token IDs)
- It impacts cost, speed and performance of models

13. Explain self attention mechanism?

Self-attention allows the model to focus on different words in the input while processing a word. It calculates the importance (weight) of every other word in the sentence to understand the context better. It helps in capturing long-range dependencies and relationships.

14. Difference between system prompt and user prompt?

System Prompt: Sets the behavior, role and rules for the AI. It gives instructions like "You are a helpful assistant".

User Prompt: Actual question or request provided by the user.

15. Elaborate the term Temperature?

Temperature is a parameter that controls randomness in the output.

- High temperature (e.g., 0.8 - 1.0): More creative, diverse output
 - Low temperature (e.g., 0 - 0.3): More focused, deterministic output
- Range: 0 to 1 (or higher in some models)

16. Classification vs Regression?

Classification	Regression
• Predicts discrete / categorical labels • Output is a class or category • Example: Spam or Not Spam, Cat or Dog • Algorithms: Logistic Regression, SVM, Decision Trees, Naive Bayes	• Predicts continuous values • Output is a real number • Example: House price, Temperature • Algorithms: Linear Regression, Random Forest Regressor



Keep Learning.

• Keep Building.

• Your Future is Intelligent!

